AUTONOMOUS STRUCTURAL SCREENING

TwinVerse Inspect AI

Inspect the unreachable.

Engineers climb bridges, rappel down dams and walk pipelines with clipboards. This does that first pass from a photograph — and tells you exactly how much to trust the answer.

Ayaan Aatif · Muhammad Muneed · Inshrah Mehmood

[Event / Track] · [Date] · Bano Qabil · Alkhidmat Foundation Pakistan

THE PROBLEM

Someone has to go and look at it.

Every bridge, dam, tunnel and building has to be inspected. Today that means a person physically getting close enough to see the concrete — climbing, rappelling, or walking the length of it with a clipboard.

It is dangerous work. It is slow enough that backlogs build and structures go years between inspections. And it is inconsistent: two inspectors looking at the same crack will often write down two different things.

The result is that most maintenance is **reactive**. Problems are found after they matter rather than before.

Dangerous	Slow	Inconsistent
People work at height and over water to look at concrete with their own eyes.	A single span can take days. Backlogs mean long gaps between inspections.	Two inspectors, two clipboards, two different answers.

WHAT WE BUILT

A first pass that never has to climb anything.

TwinVerse Inspect AI takes ordinary imagery — from a drone, a fixed camera, or a phone — and does the first pass automatically. It finds cracks, ranks how serious each one is, shows them on the photograph and on a 3D view of the structure, and produces a report anyone can forward.

The engineer still makes every decision. What changes is that they start from a sorted list of findings instead of from a memory card full of photographs.

HOW IT WORKS

Four steps, no black box.

1 · Capture	2 · Detect	3 · Score	4 · Report
Upload images or video. Drone, CCTV, robot or phone — it does not care which.	A trained model finds each crack and draws a box around it.	Each finding gets a severity score from a formula shown on screen.	Annotated images, a 3D view, and a PDF for whoever needs it.

WHY IT IS DIFFERENT

You can check its work.

Most systems of this kind give you a number and ask you to trust it. This one shows the arithmetic. Every severity score on screen is three values multiplied together, and all three are stored with the finding:

```
severity  =  area  ×  confidence  ×  class weight
0.0288  ×  0.558  ×  1.0  =  0.01606
```

A reader can recompute any figure by hand from the record it came from. That is a deliberate design choice, not a feature — it is the difference between a tool an engineer can sign off on and one they cannot.

WHERE IT STANDS

A running system, not a mockup.

104 automated tests	**6.1s** to analyse 8 images	**7ms** per image on a GPU	**84%** worst-case cracks found

Accuracy is quoted as a range rather than a number, because a single number would be a choice about which dataset to mention. On imagery resembling its training data the detector finds about **81%** of cracks. On three public datasets from unrelated sources that it had never seen, that falls to **63%**, **13%** and **8%**. All four are published in the technical record. A tool that only works on photographs like the ones it was built from is worth knowing about before it is relied on.

WHAT COMES NEXT

The honest roadmap.

Next	Later	Someday
More defect types. Tracking cracks across video frames so counts are real.	A true photogrammetric twin built from the imagery itself.	Predictive maintenance — which needs years of data that does not exist yet.

BEING HONEST ABOUT IT

We measured the weaknesses too.

Anything that inspects infrastructure has to be trustworthy before it is impressive. Every limitation below is stated inside the product itself and printed on the last page of every report it generates.

It finds cracks, and only cracks.
The model has never been shown corrosion, spalling or missing components. It will not report them even if they are in the picture.

It flags roughly three photographs in five.
That is deliberate. The threshold was set to miss as few real defects as possible, and the cost of that choice is a larger pile for a human to review. A manual inspection means looking at every photograph, after climbing the structure. The threshold is adjustable, and the trade-off is published rather than hidden.

Severity is a ranking, not a measurement.
It tells you which crack to look at first. It does not tell you how wide the crack is in millimetres — that needs camera calibration or a physical scale in the frame.

The 3D view is illustrative.
The structure shown is a generic model, and marker positions come from the order the photographs were taken, not from surveyed coordinates.

Video counts are inflated.
Frames are analysed independently, so a single crack visible across ten frames is counted ten times.

“Naming a weakness costs far less than being caught by it.”

This does not replace an engineer.

It does the first pass, so the engineer spends their time on judgement instead of data collection.

Ayaan Aatif
Team Lead

Muhammad Muneed
Team Member

Inshrah Mehmood
Team Member

TwinVerse Inspect AI

[Add contact email · repository link]